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TREND-BASED MOCK PAPER

GATE Architecture and Planning

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Architecture and Planning
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Scale 1:100, drawing 10cm -> actual:

(Numerical Answer Type — Enter the numerical value)

Section B: Architecture and Planning

Q2. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
 - (B) Differentiable only
 - (C) Continuous on $[a,b]$ and differentiable on (a,b)
 - (D) Both continuous and differentiable everywhere
-

Q3. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q4. [1-NAT] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q5. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q6. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-NAT] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
 - (B) 8
 - (C) 16
 - (D) 12
-

Q9. [1-MCQ] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q10. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q11. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
 - (B) $[2, 1]^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q13. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
 - (B) e^2
 - (C) 1
 - (D) infinity
-

Q14. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\operatorname{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q17. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
 - (B) Cosine terms
 - (C) Both
 - (D) Neither
-

Q18. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
 - (B) $1/(s-3)$
 - (C) $1/s$
 - (D) e^{-3}/s
-

Q20. [2-MCQ] [2 marks]

The differential equation $y dx - x dy = 0$ has solution:

- (A) $y = C \cdot x$
 - (B) $y = C/x$
 - (C) $y = C \cdot x^2$
 - (D) $x^2 + y^2 = C$
-

Q21. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q23. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908

- (B) 0.092
- (C) 0.762
- (D) 0.195

Q24. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q25. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q26. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q27. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q29. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q31. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q33. [1-NAT] [1 mark]

Kath-Kuni from:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Jahangir

Q35. [1-NAT] [1 mark]

M20 concrete mix ratio:

(Numerical Answer Type — Enter the numerical value)

Q36. [1-MCQ] [1 mark]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q37. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q38. [2-MCQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q39. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q40. [2-MSQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q41. [2-NAT] [2 marks]

M20 concrete mix ratio:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-NAT] [2 marks]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q43. [2-MCQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q44. [1-MCQ] [1 mark]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q45. [1-MCQ] [1 mark]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q46. [1-MCQ] [1 mark]

Cantilever UDL w , L . Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$
- (D) $wL^2/8$

Q47. [2-NAT] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle

- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q49. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q50. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q51. [2-NAT] [2 marks]

Cantilever UDL w, L. Fixed end moment:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-MCQ] [1 mark]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q53. [1-MCQ] [1 mark]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q54. [1-NAT] [1 mark]

Camber on road provides:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MSQ] [2 marks]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q56. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q58. [2-MCQ] [2 marks]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q59. [1-MCQ] [1 mark]

Trap in plumbing:

- (A) Prevents smell
- (B) Stores water
- (C) Both
- (D) None

Q60. [1-MCQ] [1 mark]

Sun path in northern hemisphere:

- (A) South
- (B) North
- (C) East
- (D) West

Q61. [1-NAT] [1 mark]

M20 concrete ratio:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Zoning regulates:

- (A) Land use
- (B) Building height
- (C) Both
- (D) None

Q63. [1-MCQ] [1 mark]

Golden ratio approximately:

- (A) 1.414
- (B) 1.618
- (C) 2.0
- (D) 3.14

Q64. [1-MCQ] [1 mark]

Load-bearing wall thickness:

- (A) 23cm
- (B) 11cm
- (C) 46cm

(D) 5cm

Q65. [1-MCQ] [1 mark]

Hardscape includes:

- (A) Plants
 - (B) Pavement
 - (C) Water
 - (D) Trees
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	B	1
3	1MCQ	B	1
4	1NAT	B	1
5	1MCQ	C	1
6	1MCQ	B	1
7	1MCQ	A	1
8	1MCQ	C	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	C	1
12	1MCQ	C	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1NAT	A	1
17	1MCQ	B	1
18	1NAT	B	1
19	1MCQ	A	1
20	1NAT	C	1
21	1NAT	C	1
22	1MCQ	A	1
23	1NAT	A	1
24	1NAT	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1NAT	B	1

29	1MCQ	A	1
30	1MCQ	B	1
31	2NAT	A	2
32	2NAT	C	2
33	2MCQ	A	2
34	2MCQ	A	2
35	2NAT	B	2
36	2MCQ	A	2
37	2MCQ	B	2
38	2NAT	B	2
39	2MCQ	A	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2MCQ	A	2

44	2NAT	B	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2NAT	A	2
48	2MCQ	C	2
49	2MCQ	C	2
50	2MCQ	A	2
51	2MCQ	A	2
52	2NAT	A	2
53	2MSQ	B	2
54	2MSQ	B	2
55	2NAT	A	2
56	2MCQ	B	2
57	2MCQ	A	2
58	2MCQ	A	2

59	2NAT	A	2
60	2NAT	A	2
61	2NAT	A	2
62	2NAT	B	2
63	2MCQ	C	2
64	2MCQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: Vernacular | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: History | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Building Materials | Topic: Concrete | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Urban Planning | Topic: Zoning | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Design | Topic: Principles | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Technology | Topic: Structures | Difficulty: moderate

Q8.

Answer: C

Solution:

Correct Answer: C

Subject: Climate & Services | Topic: Plumbing | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Climate & Services | Topic: Solar | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Landscape Architecture | Topic: Elements | Difficulty: easy

Q11.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q12.

Answer: C

Solution:

Rolle's: f continuous on [a,b], differentiable on (a,b), $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q13.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000$ MPa.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q14.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q15.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: difficult

Q16.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q17.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q18.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q19.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q22.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q23.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q43.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q46.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q47.

Answer: A

Solution:

$$DF_{AB} = 4/(4+3) = 4/7 = 0.57.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q48.

Answer: C

Solution:

$$R_A = (L-x)/L. \text{ Linear decreasing from } L \text{ to } 0.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q49.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q50.

Answer: A

Solution:

$$DF_{AB} = 4/(4+3) = 4/7 = 0.57.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q51.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q52.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q54.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q55.

Answer: A

Solution:

$$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa.}$$

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q56.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q57.

Answer: A

Solution:

$$DF_{AB} = 4/(4+3) = 4/7 = 0.57.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q58.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q60.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: A

Solution:

$$FEM = wL \cdot (L/2) = wL^2/2 \text{ (hogging).}$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q62.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q63.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q65.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 31 questions, 50 marks
Architecture: 11 questions, 18 marks
Structural Analysis: 8 questions, 13 marks
Civil Engineering: 7 questions, 11 marks
Climate & Services: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Building Materials: 1 questions, 1 marks
Urban Planning: 1 questions, 1 marks
Design: 1 questions, 1 marks
Construction Technology: 1 questions, 1 marks
Landscape Architecture: 1 questions, 1 marks

Difficulty Distribution:

Easy: 25 | Moderate: 30 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.